Oleksii Sokoliuk

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Research Interests

I'm a cosmologist, mainly interested in the complicated, high resolution simulations of the Large Scale Structure of the Universe within the non-standard theories of modified gravitation or dark matter. As well, I'm trying to alleviate H_0, σ_8 and other cosmological tensions using MG theories, find new physics in higher order weak lensing statistics.

Work Experience

Research Scholar - Main Astronomical Observatory, NAS of Ukraine 2021 - Present
– Laboratory for the Large Scale Structure of the Universe

Research Scholar - Taras Shevchenko National University of Kyiv 2019 - Present
– Kyiv Astronomical Observatory, Lisnyky Observational Station

Education

BSc (Honours) Physics - University of Aberdeen, UK Sept 2023 - Jun 2027
– Supervisor: Prof. Charles Wang

Talks & Poster Presentations

[1] **Faculty of Physics, Taras Shevchenko National University of Kyiv** 26 Apr - 30 Apr 2021
27-th Young Scientists' Conference on Astronomy and Space Physics (contributed talk)

[2] **Faculty of Physics, Odesa I.I. Mechnikov National University** 16 Aug - 20 Aug 2021
XXI Gamow International Astronomical Conference-School (contributed talk)

[3] **Faculdade de Ciências da Universidade de Lisboa** 30 May - 1 Jun 2023
CosmoVerse@Lisbon, First Annual Conference (poster)

[4] **Kobe International Conference Center, Kobe Port Island** 4 Aug - 8 Aug 2023
CCP2023 - 34th IUPAP Conference on Computational Physics (contributed talk)

[5] **Institute for Computational Cosmology, Durham University** 14 Jun 2024
Friday Lunchtime Astrophysics Talks (FLAT) (planning to attend) (invited talk)

[6] **Faculty of Physics and Astronomy, Jagiellonian University** 9 Jul - 11 Jul 2024
CosmoVerse@Krakow, Second Annual Conference (planning to attend) (poster)

Selected Publications

As of April 2024, 20 papers were published in international, peer reviewed journals with 13 papers published as a first author, 2 as a second author. In total, those papers have > 220 citations and h-index of 10 according to Google Scholar. Here are some selected publications:

[1] Oleksii Sokoliuk and Alexander Baransky. "On the existence and stability of traversable wormhole solutions in modified theories of gravity". In: *Eur. Phys. J. C* 81.8 (2021), p. 781.

[2] Oleksii Sokoliuk and Alexander Baransky. "Cosmological constraints on bulk viscous f(Q,T) gravity". In: *Astron. Nachr.* 343.5 (2022), e220003.

[3] Oleksii Sokoliuk, Alexander Baransky, Andrew Khorolskiy, et al. "An X-Ray and Optical Study of the UGSU-Type Dwarf Nova Gaia18awg". In: *Journal of Physical Studies* 26.3 (Sept. 2022), pp. 3901–3909.

[4] Oleksii Sokoliuk, Alexander Baransky, and P. K. Sahoo. "Kuchowicz gravastars in the braneworld formalism". In: *Phys. Lett. B* 829 (2022), p. 137048.

[5] Oleksii Sokoliuk, Alexander Baransky, and P. K. Sahoo. "Probing the existence of the ZTF Casimir wormholes in the framework of f(R) gravity". In: *Nucl. Phys. B* 980 (2022), p. 115845.

[6] Oleksii Sokoliuk, Alexander Baransky, and Praduymn Kumar Sahoo. "Non-singular T–K axion stars with/without the dynamical bosonic field in the presence of negative Λ term". In: *Phys. Dark Univ.* 35 (2022), p. 100972.

[7] Oleksii Sokoliuk, Zinnat Hassan, et al. "Traversable wormholes with charge and non-commutative geometry in the $f(Q)$ gravity". In: *Annals Phys.* 443 (2022), p. 168968.

[8] Oleksii Sokoliuk, Sanjay Mandal, et al. "Generalised Ellis–Bronnikov wormholes in $f(R)$ gravity". In: *Eur. Phys. J. C* 82.4 (2022), p. 280.

[9] Oleksii Sokoliuk, Sneha Pradhan, et al. "Buchdahl quark stars within $f(Q)$ theory". In: *Eur. Phys. J. Plus* 137.9 (2022), p. 1077.

[10] Oleksii Sokoliuk, Subhrat Praharaj, et al. "Accretion flows around exotic tidal wormholes - I. Ray-tracing". In: *Astron. Astrophys.* 665 (2022), A139.

[11] Sanjay Mandal, Oleksii Sokoliuk, et al. "H_0 tension in torsion-based modified gravity". In: *Nucl. Phys. B* 993 (2023), p. 116285.

[12] Oleksii Sokoliuk, Simran Arora, et al. "On the impact of $f(Q)$ gravity on the large scale structure". In: *Mon. Not. Roy. Astron. Soc.* 522.1 (2023), pp. 252–267.

[13] Oleksii Sokoliuk, Alexander Baransky, and P. K. Sahoo. "Compact stars admitting Finch-Skea symmetry in the presence of various matter fields". In: *Chin. Phys. C* 47.1 (2023), p. 015104.

[14] Fabiano F. Santos, Oleksii Sokoliuk, and Alexander Baransky. "Holographic Complexity of Braneworld in Horndeski Gravity". In: *Fortschritte der Physik* 71.2-3 (2023), p. 2200141.

[15] Fabiano F. Santos (including Oleksii Sokoliuk) et al. "AdS/BCFT Correspondence and Horndeski Gravity in the Presence of Gauge Fields: Holographic Paramagnetism/ Ferromagnetism Phase Transition". In: *Fortschritte der Physik* 71.12 (2023), p. 2300008.

[16] Oleksii Sokoliuk et al. "AdS Black Hole Thermodynamics and Microstructures from $f(Q)$ Gravitation". In: *Fortschritte der Physik* 72.1 (2024), p. 2300043.

Observations of near-earth objects and comets were published in 12 Minor Planet Electronic Circulars (MPEC 🔗) and observations of Gamma Ray Bursts in GRB Coordinates Network (GCN 🔗).

Collaborative Papers

Additionally, within the GRANDMA collaboration, I actively contributed to various publications, engaging in tasks such as observations, data analysis, figure/table creation, and text revision:

[1] V. Aivazyan (including Oleksii Sokoliuk) et al. "GRANDMA observations of ZTF/Fink transients during summer 2021". In: *Mon. Not. Roy. Astron. Soc.* 515.4 (2022), pp. 6007–6022.

[2] D. A. Kann (including Oleksii Sokoliuk) et al. "GRANDMA and HXMT Observations of GRB 221009A: The Standard Luminosity Afterglow of a Hyperluminous Gamma-Ray Burst—In Gedenken an David Alexander Kann". In: *Astrophys. J. Lett.* 948.2 (2023), p. L12.

[3] T Hussenot-Desenonges (including Oleksii Sokoliuk) et al. "Multi-band analyses of the bright GRB 230812B and the associated SN2023pel". In: *Mon. Not. Roy. Astron. Soc.* (2024), stae503.

[4] I. Tosta e Melo (including Oleksii Sokoliuk) et al. "Ready for O4 II: GRANDMA observations of Swift GRBs over eight weeks in spring 2022". In: *Astron. Astrophys.* 682 (2024), A141.

Conference Proceedings & Other

[1] S. Agayeva (including Oleksii Sokoliuk) et al. "The GRANDMA network in preparation for the fourth gravitational-wave observing run". In: *Observatory Operations: Strategies, Processes, and Systems IX* 12186 (Aug. 2022), 121861H.

In Preparation & Under Review

[1] Jialian Liu (including Oleksii Sokoliuk) et al. "Early-Time Observations of SN 2023wrk: A Luminous Type Ia Supernova with Significant Unburned Carbon in the Outer Ejecta". In: *Astrophys. J.* (2024).

[2] Oleksii Sokoliuk. "Probing Physics beyond ΛCDM with JWST data up to $z \sim 17$". In: *Astron. Astrophys.* (2024).

[3] Oleksii Sokoliuk et al. "The `LANCELOT` project: Cosmological simulations for Large Scale Structure in the modified theories of gravitation with massive neutrinos". In: *Mon. Not. Roy. Astron. Soc.* (2024).

Peer Review

- European Physical Journal C (2022)
- New Astronomy (2022)
- Foundations of Physics (2022)
- Scientific Reports (2023)
- Annalen der Physik (2023)
- Indian Journal of Physics (2023)
- Physics of the Dark Universe (2023 (2))
- Pramana (2024)
- Annals of Physics (2024 (2))

Memberships

- International Society for Relativistic Quantum Information (ISRQI)
- International Society for Quantum Gravity (ISQG)
- Global Rapid Advanced Network Devoted to the Multi-messenger Addicts (GRANDMA)
- Assembling Galaxies Of Resolved Anatomy (AGORA)
- Insitute of Physics (IoP) Associate Member
- American Astronomical Society (AAS) Undergraduate Student Member

Awards & Grants

Monetary awards (6200$ total):

- Scholarship of the President of Ukraine - 1000$
- CA21136 ITC Conference Grant for CCP2023 - 2200$
- CA21136 Conference Grant for CosmoVerse@Lisbon - 1000$
- CA21136 Conference Grant for CosmoVerse@Krakow - 1000$ (applied)
- MAO NAS travel grant - 2 × 500$

Computing time allocations (1.65M CPUh total):

- PI: 1.15M CPUh on IUCAA Pegasus, with Prof. Jiajun Zhang and Prof. P. K. Sahoo as Co-PIs (applied)
- Co-PI: ∼200k CPUh on Australian NCI Gadi with Subhrat Praharaj as PI
- Co-PI: ∼300k CPUh on IUCAA Pegasus with Prof. P. K. Sahoo as PI

Observational proposals:

- PI: Target of Opportunity (TOO) observation of Gaia18awg by SWIFT space telescope (ID: 13502, ∼3k seconds)
- PI: Target of Opportunity (TOO) observation of Gaia18awg by XMM-Newton space telescope (ID: 08711910011, ∼18k seconds)

Public Outreach

- Short article *"Is Modified Gravity an Illusion?"* for CosmoVerse COST action in a series of articles for general public "Learn about Cosmology"
- Interview for CosmoVerse COST action in a "Meet our scientists" dissemination activity
- Volunteer for Astronomy night under "Curiosity in Action" program (funded by STFC)
- STEM Ambassador Scotland, volunteering to promote science to the general public

Teaching

ST1506 - *Understanding Data* — Teaching Assistant, 2024-2025
MA1005 - *Calculus I* — Teaching Assistant, 2024-2025

Skills

Languages	Ukrainian (native tongue), Russian (native tongue), English (C1, very fluent)
Programming Languages	Python (advanced), LaTeX(advanced), Mathematica (advanced), C/C++ (intermediate), Matlab (intermediate), Haskell (beginner)
Software	**N-body/SPH/Lattice simulations:** Gadget-2/4, ME/MP/MG/Axion-Gadget, SWIFT, zwindstroom, NGenIC, 2LPTic, monofonIC, MUSIC, $\mathcal{C}$osmo$\mathcal{L}$attice **Approximate simulations, emulators:** COLA, MG-PICOLA, LPICOLA, forge. **Post-processing:** eht-imaging, UFalcon, DeepSphere, GLAMER, eMaNu, swiftsimio, VELOCIRaptor/ROCKSTAR, SAGE/RSAGE/cifog, L-Galaxies. **GRMHD, ray-tracing:** iharmd3D, athena++, BHAC, RAPTOR, gyoto. **Cosmology, MCMC:** CAMB/MGCAMB, CLASS/MG-CLASS I/II, reps, pymc3, emcee, Cobaya/MGCobaya. **Parallel Computing:** MPI, OpenMP, Slurm, worked with Cray/ARM

Projects

CA21106 - COSMIC WISPers in the Dark Universe Oct 2022 - Oct 2026
– Working Group 2: WISPs Dark Matter and Cosmology

CA21136 - CosmoVerse Oct 2022 - Oct 2026
– Working Group 3: Fundamental Physics

Mathematical Modeling in Interdisciplinary Research of Processes and Systems Based on Intelligent Supercomputer, Grid and Cloud Technologies Jan 2021 - Jan 2025
– Grant for the Lab of LSS, MAO NAS of Ukraine

Developed Software & Simulations

`RAMA`	Code can be used with `reps` output to find z, l at which N-body simulation coincide with the desired cosmology in the presence of massive neutrinos. Available at RAMA.
`MG-SWIFT`	Modification of the standard N-body/SPH SWIFT code to include such cosmologies as $f(Q)/f(T)$ gravitation, Palatini-$f(R)$ gravitation with the help of linearly interpolated Hubble parameter and effective gravitational constant, which are updated every step using provided data from tables. Available at MG-SWIFT.
`zwindstroom`	With the help of Willem Elbers, modified initial conditions generator monofonIC to include massive neutrinos and non-standard cosmologies simultaneously. Available at MG-monofonic.
`LANCELOT`	A suite of 12 high resolution N-body simulations with more than 13.5 billion particles in total as well as 24TB of output. From the simulation snapshots, many quantities such as void/halo catalogues, power spectrum and HMF are being derived. Currently in development.
`Condor`	Emulator of non-linear $P(k)$ for Jordan-Brans-Dicke cosmology, based on 200 Comoving Lagrangian Approximation simulations of structure formation with Latin hypercube parameter sampler. Currently in development.
`Anthology of Inflation`	A project, aimed to study 27 single-parameter models of inflation using $N = 256$ lattice simulations of reheating, primordial power spectrum of curvature perturbations and gravitational waves. Currently in development, some of the code is available via the link ASPIC.

References

Name	**Dr. P. K. Sahoo**	Name	**Dr. Irina Vavilova**
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